北京郵電大學

Queen Mary University of London

For examiners' use only

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Total	

EBU5212 A

Joint Programme Examinations 2023/24

EBU5212 Communication and Networks

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: Any Electronic devices including electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Yan SUN, Dr Habiba Akter

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Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) **Figure 1** shows a form of Huffman tree. Determine the code efficiency of this Huffman tree.

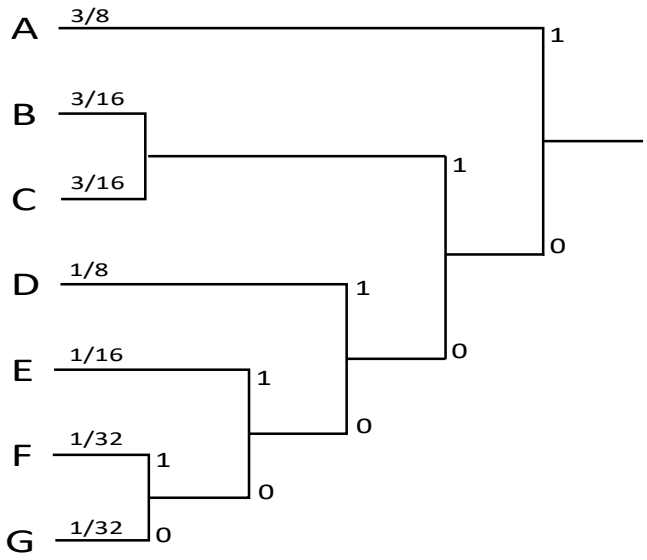

Figure 1

[12 marks]

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12 marks

b) A systematic linear block code with the following parity check matrix H is employed to protect information against errors:

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

[13 marks]

i) Find all the codewords via G matrix (generator matrix).

(8 marks)

ii) Decode the following received sequence r = 1111010.

(5 marks)

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	13 marks

Question marking: $\frac{}{12} + \frac{}{13} = \frac{}{25}$

Question 2

a) What is the "hidden terminal" problem? Provide your answer with the aid of illustration.

[7 marks]

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	7 marks

b) Describe how host computer 192.168.1.2 is able to deliver Internet Protocol datagrams to host 192.168.1.4 on the same Ethernet network, via bridge BR, illustrating your answer with reference to **Figure 2a**. Your answer should include a full description of the role played by the Address Resolution Protocol (ARP) and assume that no previous communication has taken place for some time.

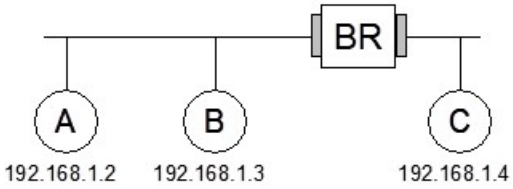

Figure 2a

[10 marks]

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		10 marks

c) The bridge is now replaced with a router RT, as shown in **Figure 2b**. Explain any differences that would arise in the process of how host 192.168.1.2 delivers data to host 192.168.2.4. Once again, assume that no previous communication has taken place for some time and the subnet mask is /24.

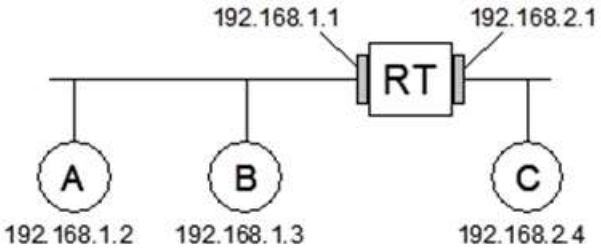

Figure 2b

[8 marks]

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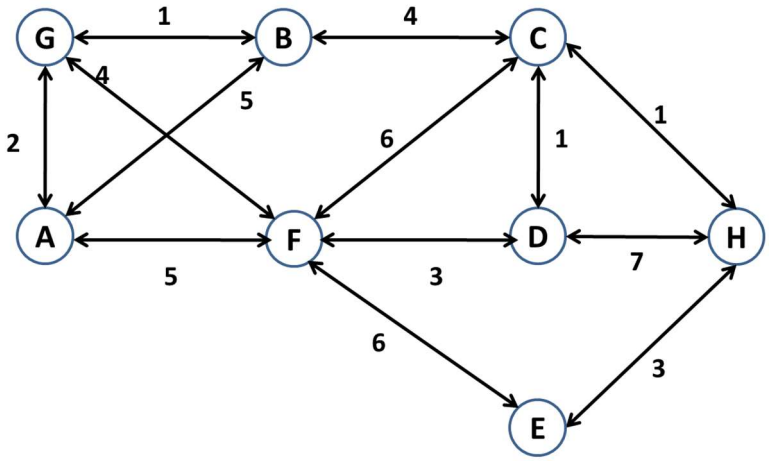
	8 marks

Question marking: $\frac{}{7} + \frac{}{10} + \frac{}{8} = \frac{}{25}$

Question 3

a) Describe how would you determine the least-cost tree for the topology below from node D using Dijkstra's algorithm. You should also illustrate the final tree topology.



[12 marks]

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12 marks

b) Applying the Distance Vector Algorithms to the topology in a), produce the routing table at node F.

[7 marks]

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	7 marks

c) Produce the updated routing table at node F with Distance Vector algorithm when node D is disconnected with node F and node G moves away from the topology in b).

[6 marks]

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6 marks

Question marking: $\frac{\quad}{12} + \frac{\quad}{7} + \frac{\quad}{6} = \frac{\quad}{25}$

Question 4

a) Considering the traditional TCP in wired IP network,

[14 marks]

i) What are the advantages and disadvantages of the fact that TCP acknowledgements are cumulative?

(9 marks)

ii) What can be deduced from the reception of duplicate acknowledgements?

(2 marks)

iii) How is this information used in TCP Reno to maintain channel utilisation?

(3 marks)

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		14 marks

b) TCP congestion control uses 4 basic mechanisms known as Additive Increase (AI), Multiplicative Decrease (MD), Slow Start and Fast Recovery. Briefly explain each of those 4 basic mechanisms with figure illustration.

[11 marks]

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		11 marks

Question marking: $\frac{}{14} + \frac{}{11} = \frac{}{25}$

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Rough Working

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